

TACTICAL RISK INDEX

AI-Powered Financial Risk Signals
for Dynamic Index Rebalancing

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ROLL NO.

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5**

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Turning financial information into explainable portfolio action.

RISK → DECISION

NEWS

Unstructured

AI

Structured

RISK

Scored

INDEX

Rebalanced

Why tactical index rebalancing matters

The core idea: portfolio weights should respond to meaningful, explainable changes in risk information.

CONCEPT

A tactical index is not a static list of stocks.

Its allocation can be adjusted when new information changes the perceived short-term risk or opportunity of its constituents.

- Market-moving information arrives continuously through news and public sources.
- The same headline can have different implications depending on the company and event type.
- Rebalancing converts this information into a controlled change in portfolio weights.
- The objective is not to replace investment judgment, but to create a repeatable decision-support mechanism.

THE DECISION LOOP

- 01 Observe information
- 02 Interpret the event
- 03 Estimate materiality
- 04 Adjust allocation

How the AI risk signal is constructed

Natural-language financial information is converted into a structured signal that downstream allocation logic can consume.

SENTIMENT

What is the direction of the information?

Positive / Negative / Neutral
Score: -1 to +1

EVENT TYPE

What kind of risk or catalyst is present?

M&A • Macro • Cyber
Product • Labor • Geopolitical

IMPACT

How material could the event be?

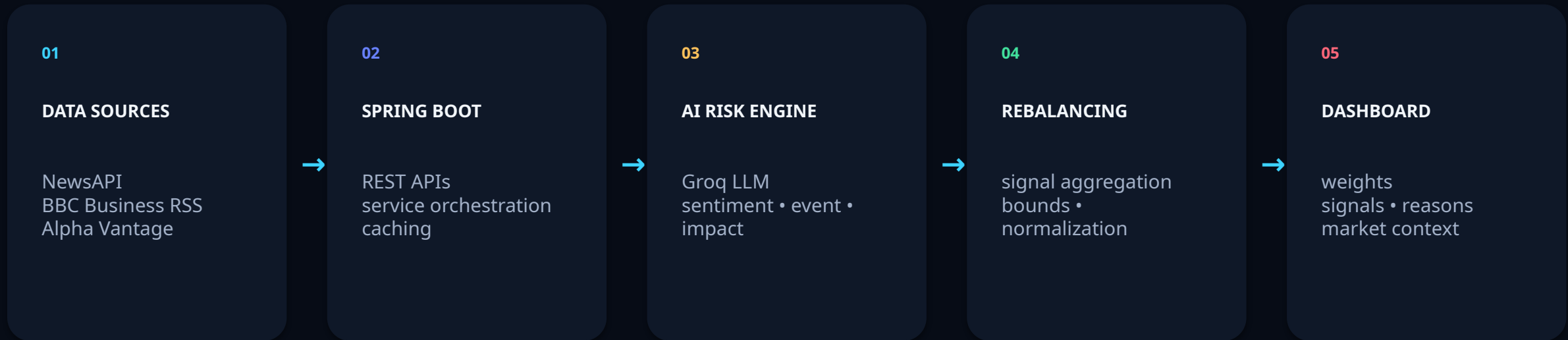
Score: 1 to 10
Higher = stronger allocation influence

AFFECTED STOCKS

The model maps company/product references to a controlled ticker universe — e.g. NVIDIA → NVDA, Apple → AAPL, Microsoft/Azure → MSFT.

System architecture: from raw information to portfolio action

Each layer has a focused responsibility so ingestion, intelligence and allocation can evolve independently.



DESIGN PRINCIPLE • raw text is never sent directly into allocation — it becomes a validated, explainable RiskSignal first.

Rebalancing logic: turning risk into a constrained weight change

A transparent rule keeps the prototype explainable while preventing extreme allocations.

$$\text{Weight Adjustment} = \text{Sentiment} \times \text{Impact Multiplier} \times \text{Rebalancing Factor}$$

SIGNAL AGGREGATION

Multiple signals affecting the same stock are combined using sentiment and impact so stronger events carry more influence.

RISK BOUNDS

Every stock is constrained to a minimum and maximum weight to prevent a single headline from dominating the index.

NORMALIZATION

After individual adjustments, weights are normalized so the final portfolio remains exactly 100%.

Interpretation: positive, high-impact signals tend to increase exposure; negative, high-impact signals tend to reduce exposure.

What the prototype demonstrates

The value is the complete decision chain: information → intelligence → allocation → explanation.

10

Stocks in tactical universe

100%

Final normalized allocation

5

Risk signals in demo

3

Public data/API sources

WHY THIS IS USEFUL

- Creates a consistent framework for interpreting financial headlines.
- Makes the allocation decision traceable to an underlying risk signal.
- Provides both the new weight and an AI-generated reason.
- Separates data ingestion, AI interpretation and portfolio logic.

EXAMPLE OUTPUT

AAPL 10.00% → **11.62%**

Weight movement is bounded and normalized with the rest of the index.

Limitations, roadmap and the bigger picture

The prototype prioritizes clarity and explainability; production scale would require stronger validation and historical evaluation.

CURRENT PROTOTYPE

10-stock universe
headline-driven signals
external APIs
rule-based allocation

NEXT ITERATION

Historical backtesting
signal persistence
more data sources
portfolio optimization

PRODUCTION DIRECTION

AWS deployment
observability
scalable event pipeline
stronger financial models

THE CORE IDEA

Unstructured financial information → structured risk intelligence → explainable tactical allocation

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